Take-Home Exercise: Instrument Price Dashboard

React + C#/.NET •

Overview

You've been given a CSV of synthetic daily closing prices for 200 instruments over a 30-day window (columns: date, ticker, price). Build a small full-stack app that lets a user browse instruments, view a price chart, and see basic computed statistics.

What to build

Backend (C#)

Load the provided CSV (in-memory is fine — no database required) and expose:

Method	Route	Returns
GET	/api/instruments	List of all 200 tickers
GET	/api/prices/{ticker}	Full 30-day time series for one ticker (404 if unknown)
GET	/api/prices/{ticker}/stats	Computed: total return %, daily volatility (stdev of returns), max drawdown

Stats definitions (feel free to note your own interpretation in the README if you go a different direction):

- Total return % = (last price / first price – 1) × 100
- Daily volatility = standard deviation of daily returns over the window
- Max drawdown = largest peak-to-trough decline over the window

Frontend (React)

- A list/search view of all 200 tickers (search-as-you-type is fine, no need for fuzzy matching)
- Selecting a ticker shows a price line chart over the 30-day window (any charting library)
- Display the computed stats (return %, volatility, max drawdown) alongside the chart
- Handle loading and error states (e.g., API unreachable, unknown ticker)
- Multi-select 2–3 tickers and overlay them on a single chart for comparison

Constraints & ground rules

- This should run locally
- You may use any charting library (Recharts, Chart.js, Victory, etc.) and any HTTP client
- AI coding assistants are fine to use — note in your README where they helped

What to submit

- A git repo (zip or link) with backend and frontend